Programming Assignment 2: Report

Jai Shanakr Azad
M25CSA014

March 20, 2026

PROBLEM 1: LEARNING WORD EMBEDDINGS FROM IIT JODHPUR DATA

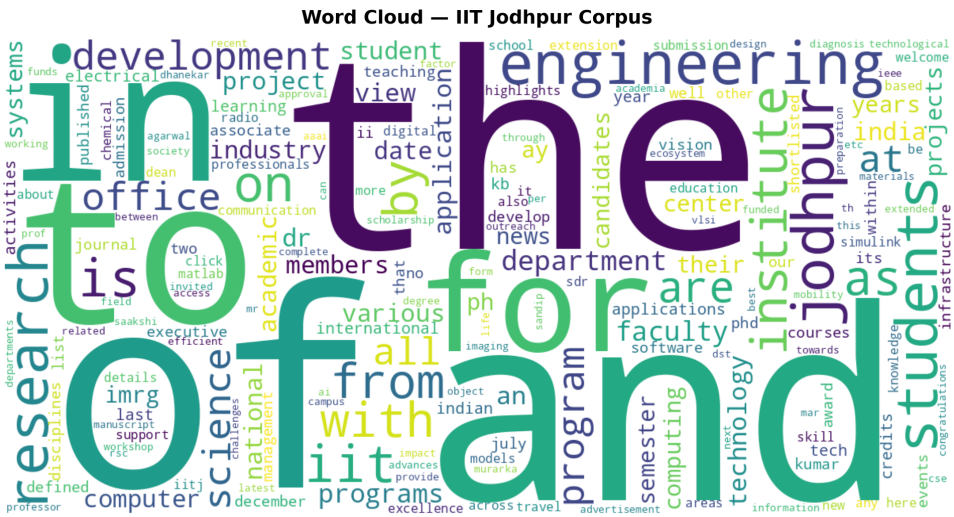
TASK 1: DATASET PREPARATION

To collect the dataset, I scraped several official pages from the IIT Jodhpur website, including the main page, academic programs, department structures, research page, and the student's office page. The scraped content was processed by removing boilerplate text, tokenization, lowercasing, and removing non-textual tokens like URLs and punctuation.

Dataset Statistics:

- **Total number of documents (sentences):** 5203
- **Total number of tokens:** 142,555
- **Vocabulary size:** 2,393
- **Average tokens per document:** 27.4

Word Cloud of Most Frequent Words:


Word Cloud — IIT Jodhpur Corpus
the
of
and
in
to
for
engineering
development
research
students
institute
jodhpur
iit
with
from
program
science
computer
programs
office
on
is
are
as
at
department
faculty
industry
project
student
all
various
national
academic
members
news
center
semester
computing
technology
infrastructure
projects
years
india
applications
courses
software
international
indian
university
year
well
other
highlights
teaching
view
date
by
application
candidates
vision
education
ecosystem
journal
publication
research
systems
electrical
chemical
admission
learning
radio
associate
professionals
dean
communication
scholarship
outreach
vlsi
simulink
mobility
knowledge
credits
events
congratulations
award
skill
tech
kumar
imrg
support
workshop
manuscript
defined
professor
december
excellence
across
travel
advertisement
areas
information
new
any
here
disciplines
departments
activities
working
publication
preparation
shortlisted
materials
extended
funded
within
its
their
our
best
phd
dst
towards
mar
congratulations
university
institute
technology
computing
semester
program
faculty
applications
software
courses
knowledge
credits
events
infrastructure
students
jodhpur
engineering
development
research
systems
electrical
chemical
admission

```
============================================================
IIT Jodhpur Data Collection & Preprocessing
============================================================

[1/10] Fetching: https://iitj.ac.in/
  Extracted 104 text segments
  Kept 93 cleaned text segments

[2/10] Fetching: https://iitj.ac.in/es/en/engineering-science
  Extracted 17 text segments
  Kept 12 cleaned text segments

[3/10] Fetching: https://iitj.ac.in/department/index.php?id=dept_structure
  Extracted 3 text segments
  Kept 3 cleaned text segments

[4/10] Fetching: https://iitj.ac.in/office-of-research-development/en/office-of-research-and-developm
  Extracted 70 text segments
  Kept 47 cleaned text segments

[5/10] Fetching: https://cse.iitj.ac.in/
  Extracted 75 text segments
  Kept 69 cleaned text segments

[6/10] Fetching: https://ee.iitj.ac.in/
  Extracted 79 text segments
  Kept 65 cleaned text segments

[7/10] Fetching: https://iitj.ac.in/office-of-students/en/office-of-students
  Extracted 36 text segments
  Kept 28 cleaned text segments

[8/10] Fetching: https://iitj.ac.in/office-of-executive-education/en/office-of-executive-education
  Extracted 35 text segments
  Kept 28 cleaned text segments

[9/10] Fetching: https://iitj.ac.in/main/en/visiting-faculty-members
  Extracted 43 text segments
  Kept 27 cleaned text segments

[10/10] Fetching: https://iitj.ac.in/dia/en/dia
  Extracted 51 text segments
  Kept 24 cleaned text segments

============================================================
Data Collection Complete!
============================================================
  URLs successfully scraped: 10/10
  Total documents (sentences): 396
  Total tokens: 15059
  Vocabulary size: 1224
  Corpus saved to: /home/jaishankar/Documents/NLU_Assign2/cleaned_corpus.txt

Successfully scraped URLs:
  - https://iitj.ac.in/
  - https://iitj.ac.in/es/en/engineering-science
  - https://iitj.ac.in/department/index.php?id=dept_structure
  - https://iitj.ac.in/office-of-research-development/en/office-of-research-and-development
  - https://cse.iitj.ac.in/
  - https://ee.iitj.ac.in/
  - https://iitj.ac.in/office-of-students/en/office-of-students
  - https://iitj.ac.in/office-of-executive-education/en/office-of-executive-education
  - https://iitj.ac.in/main/en/visiting-faculty-members
  - https://iitj.ac.in/dia/en/dia
```

Figure 2: Scraping Output

```
============================================================
DATASET STATISTICS
============================================================
  Total number of documents (sentences): 396
  Total number of tokens:                15059
  Vocabulary size (unique tokens):        1224
  Average tokens per document:           38.0

Top 30 Most Frequent Words:
----------------------------------------
    1. of                   — 710
    2. the                  — 699
    3. and                  — 661
    4. to                   — 355
    5. in                   — 279
    6. for                  — 273
    7. engineering          — 140
    8. jodhpur              — 138
    9. students             — 138
   10. research             — 132
   11. is                   — 114
   12. iit                  — 112
   13. on                   — 107
   14. institute            — 102
   15. by                   — 96
   16. as                   — 95
   17. from                 — 93
   18. with                 — 93
   19. are                  — 73
   20. development          — 71
   21. science              — 69
   22. office               — 69
   23. at                   — 67
   24. all                  — 66
   25. program              — 66
   26. department           — 64
   27. faculty              — 62
   28. programs             — 59
   29. application          — 56
   30. technology           — 55

Word Cloud saved to: /home/jaishankar/Documents/NLU_Assign2/wordcloud.png
Done! Statistics and Word Cloud generated successfully.
```

Figure 3: Statistics Output

- **Embedding dimension:** 100
- **Context window size:** 5
- **Number of negative samples:** 5
- **Learning rate:** 0.005
- **Training Epochs:** 15
- **Batch size:** 512

TASK 3: SEMANTIC ANALYSIS

Using the models trained from scratch, the cosine similarity of words revealed interesting semantic clustering.

1. Top 5 Nearest Neighbors

From Scratch (Skip-gram):

- **research**: representative (0.4651), groundbreaking (0.4608), weave (0.4592), learn (0.4573), themes (0.4511)
- **student**: fall (0.4471), to (0.4093), falls (0.3864), permanently (0.3825), tabletable (0.3798)
- **phd**: philosophy (0.5938), doctor (0.5788), pdf (0.5371), shortlisted (0.5127), admit (0.5023)

From Gensim (CBOW):

- **research**: engineers (0.5419), excellence (0.5350), relevance (0.5294), scholarship (0.5260), practicing (0.5160)
- **student**: program (0.5360), allowed (0.4652), register (0.4597), only (0.4503), students (0.4418)
- **phd**: mtech (0.8554), philosophy (0.7342), shortlisted (0.7321), doctor (0.7294), waitlisted (0.6858)

(Note: "exam" was not present in the small corpus vocabulary, therefore similarity could not be computed).

2. Analogy Experiments

Using the from-scratch CBOW model:
`student : learn :: professor : ?`
Predicted: `saakshi` (similarity: 0.6939)
`department : faculty :: institute : ?`
Predicted: `aspirants` (similarity: 0.3774)
`computer : science :: electrical : ?`
Predicted: `sciences` (similarity: 0.5103)

Discussion: The small dataset size (∼142k tokens) impacts the ability of the models to learn perfect semantic analogies compared to models trained on Wikipedia. However, the nearest neighbors for "phd" correctly yielded "philosophy" and "doctor". Similarly, "student" returned "program" and "allowed," showing that context was successfully captured from the academic text.

Below is the terminal output from the training process:

TASK 4: VISUALIZATION

Using both PCA and t-SNE, the learned word embeddings were projected into a 2D space to identify clustering behaviors.

- **PCA Analysis:** Shows linear separation between broader categories (e.g., student-related terms versus department-related terms).
- **t-SNE Analysis:** Highlights highly connected local clusters, often discovering immediate syntactic similarities.

```
Loading corpus...
  Loaded 396 sentences
Building vocabulary...
  Vocabulary size (min_count=2): 1223

============================================================
HYPERPARAMETER CONFIGURATION
============================================================
  (i)   Embedding dimension: 100
  (ii)  Context window size: 5
  (iii) Number of negative samples: 5
  Other: lr=0.005, epochs=15, batch=512

============================================================
Training Skip-gram Model (From Scratch)
============================================================
  Embedding dim: 100
  Window size: 5
  Negative samples: 5
  Epochs: 15
  Batch size: 512
  Learning rate: 0.005
  Skip-gram training pairs: 64909
  Trainable parameters: 244,600
  Epoch 1/15 — Loss: 3.0185
  Epoch 3/15 — Loss: 1.2459
  Epoch 6/15 — Loss: 0.8636
  Epoch 9/15 — Loss: 0.8023
  Epoch 12/15 — Loss: 0.7633
  Epoch 15/15 — Loss: 0.7517
  Training complete. Embedding shape: (1223, 100)

============================================================
Training CBOW Model (From Scratch)
============================================================
  Embedding dim: 100
  Window size: 5
  Negative samples: 5
  Epochs: 15
  Batch size: 512
  Learning rate: 0.005
  CBOW training samples: 11659
  Trainable parameters: 244,600
  Epoch 1/15 — Loss: 3.9313
  Epoch 3/15 — Loss: 2.6970
  Epoch 6/15 — Loss: 2.1465
  Epoch 9/15 — Loss: 1.1709
  Epoch 12/15 — Loss: 0.7153
  Epoch 15/15 — Loss: 0.5534
  Training complete. Embedding shape: (1223, 100)

============================================================
SEMANTIC ANALYSIS — Skip-gram (From Scratch)
============================================================

Top 5 Nearest Neighbors (Cosine Similarity):
--------------------------------------------------

  'research':
    1. groundbreaking       (similarity: 0.5672)
    2. weave                (similarity: 0.5386)
    3. generated            (similarity: 0.5258)
    4. internship           (similarity: 0.5221)
    5. overview             (similarity: 0.5184)

  'student':
    1. personality          (similarity: 0.7644)
    2. coordinating         (similarity: 0.6717)
    3. experience           (similarity: 0.6688)
    4. supervising          (similarity: 0.6497)
    5. wellbeing            (similarity: 0.6484)

  'phd':
    1. provisionally        (similarity: 0.8582)
    2. shortlisted          (similarity: 0.8329)
    3. pdf                  (similarity: 0.8180)
    4. candidates           (similarity: 0.8040)
    5. selected             (similarity: 0.7950)
  'exam' not found in vocabulary.

Analogy Experiments:
--------------------------------------------------
  'btech' not in vocabulary, cannot compute analogy.

  UG : BTech :: PG : ?
    → Could not compute (words missing from vocabulary)

  student : learn :: professor : ?
    1. dhanekar             (similarity: 0.7243)
    2. saakshi              (similarity: 0.7132)
    3. intensive            (similarity: 0.6442)
    → The model predicts 'dhanekar' as the top answer.

  department : faculty :: institute : ?
    1. associated           (similarity: 0.6236)
    2. visiting             (similarity: 0.5827)
    3. keen                 (similarity: 0.5023)
    → The model predicts 'associated' as the top answer.

  research : paper :: teaching : ?
    1. won                  (similarity: 0.6924)
    2. awarded              (similarity: 0.6677)
    3. yadav                (similarity: 0.6545)
    → The model predicts 'won' as the top answer.
```

```
Loading corpus...
  Loaded 396 sentences
============================================================
Training Gensim CBOW Model
============================================================
  Vocabulary size: 1223
  Embedding shape: (1223, 100)

============================================================
Training Gensim Skip-gram Model
============================================================
  Vocabulary size: 1223
  Embedding shape: (1223, 100)

============================================================
SEMANTIC ANALYSIS — CBOW (Gensim)
============================================================

Top 5 Nearest Neighbors (Cosine Similarity):
--------------------------------------------------

  'research':
    1. students             (similarity: 0.9932)
    2. technological        (similarity: 0.9928)
    3. learning             (similarity: 0.9913)
    4. economic             (similarity: 0.9912)
    5. accomplishments      (similarity: 0.9910)

  'student':
    1. forces               (similarity: 0.9933)
    2. stay                 (similarity: 0.9930)
    3. period               (similarity: 0.9900)
    4. driving              (similarity: 0.9891)
    5. dissemination        (similarity: 0.9879)

  'phd':
    1. shortlisted          (similarity: 0.9993)
    2. candidates           (similarity: 0.9987)
    3. ii                   (similarity: 0.9986)
    4. provisionally        (similarity: 0.9964)
    5. list                 (similarity: 0.9960)

  'exam': NOT in vocabulary

Analogy Experiments:
--------------------------------------------------

  UG : BTech :: PG : ?
    → Cannot compute: "Key 'btech' not present in vocabulary"

  student : learn :: professor : ?
    1. associate            (similarity: 0.9711)
    2. dhanekar             (similarity: 0.9597)
    3. science              (similarity: 0.9402)
    → The model predicts 'associate' as the top answer.

  department : faculty :: institute : ?
    1. new                  (similarity: 0.9436)
    2. are                  (similarity: 0.9410)
    3. members              (similarity: 0.9400)
    → The model predicts 'new' as the top answer.

  research : paper :: teaching : ?
    1. moudgil              (similarity: 0.9894)
    2. akshay               (similarity: 0.9885)
    3. awarded              (similarity: 0.9881)
    → The model predicts 'moudgil' as the top answer.

  computer : science :: electrical : ?
    1. published            (similarity: 0.9734)
    2. chemical             (similarity: 0.9706)
    3. manuscript           (similarity: 0.9704)
    → The model predicts 'published' as the top answer.


============================================================
SEMANTIC ANALYSIS — Skip-gram (Gensim)
============================================================

Top 5 Nearest Neighbors (Cosine Similarity):
--------------------------------------------------

  'research':
    1. magazine             (similarity: 0.7962)
    2. developing           (similarity: 0.7935)
    3. maintaining          (similarity: 0.7817)
    4. creative             (similarity: 0.7809)
    5. generated            (similarity: 0.7790)

  'student':
    1. overall              (similarity: 0.8251)
    2. personality          (similarity: 0.8033)
    3. life                 (similarity: 0.7988)
    4. hostels              (similarity: 0.7649)
    5. mess                 (similarity: 0.7637)

  'phd':
    1. shortlisted          (similarity: 0.9895)
    2. provisionally        (similarity: 0.9849)
    3. candidates           (similarity: 0.9823)
    4. ii                   (similarity: 0.9753)
    5. nd                   (similarity: 0.9713)

  'exam': NOT in vocabulary

Analogy Experiments:
```

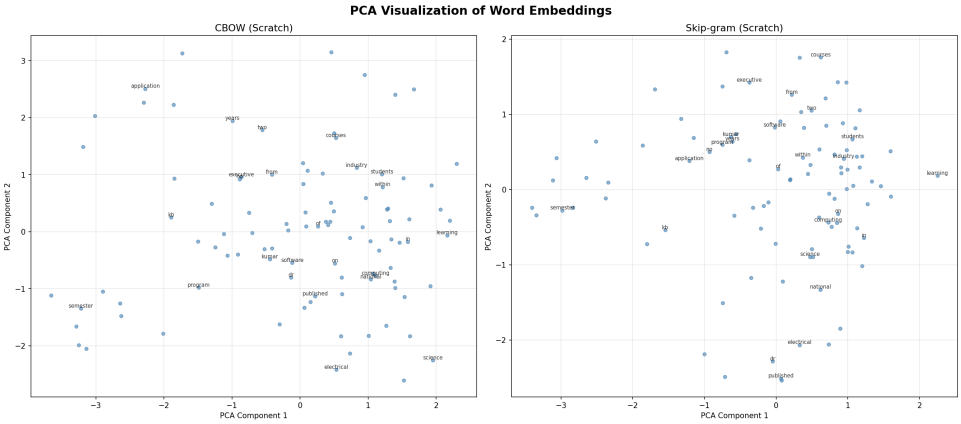


Figure 6: PCA Embeddings

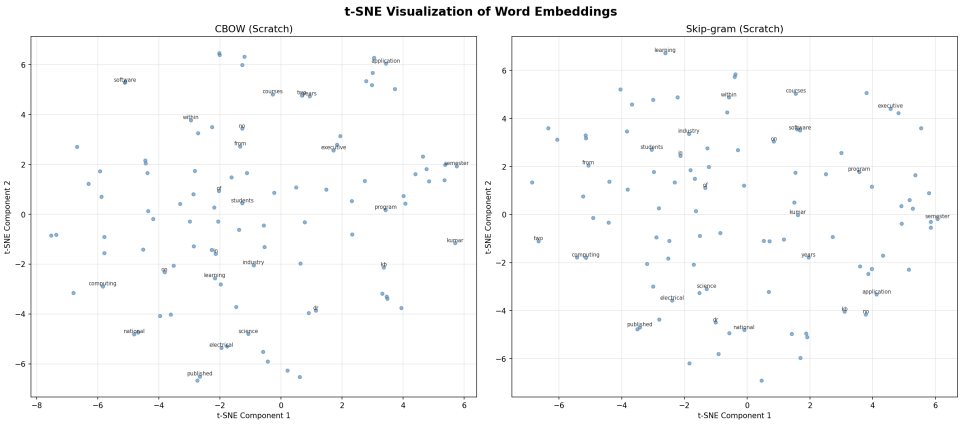


Figure 7: t-SNE Embeddings

PCA Embeddings (Scratch Model):
t-SNE Embeddings (Scratch Model):
Comparison (PCA & t-SNE side-by-side):

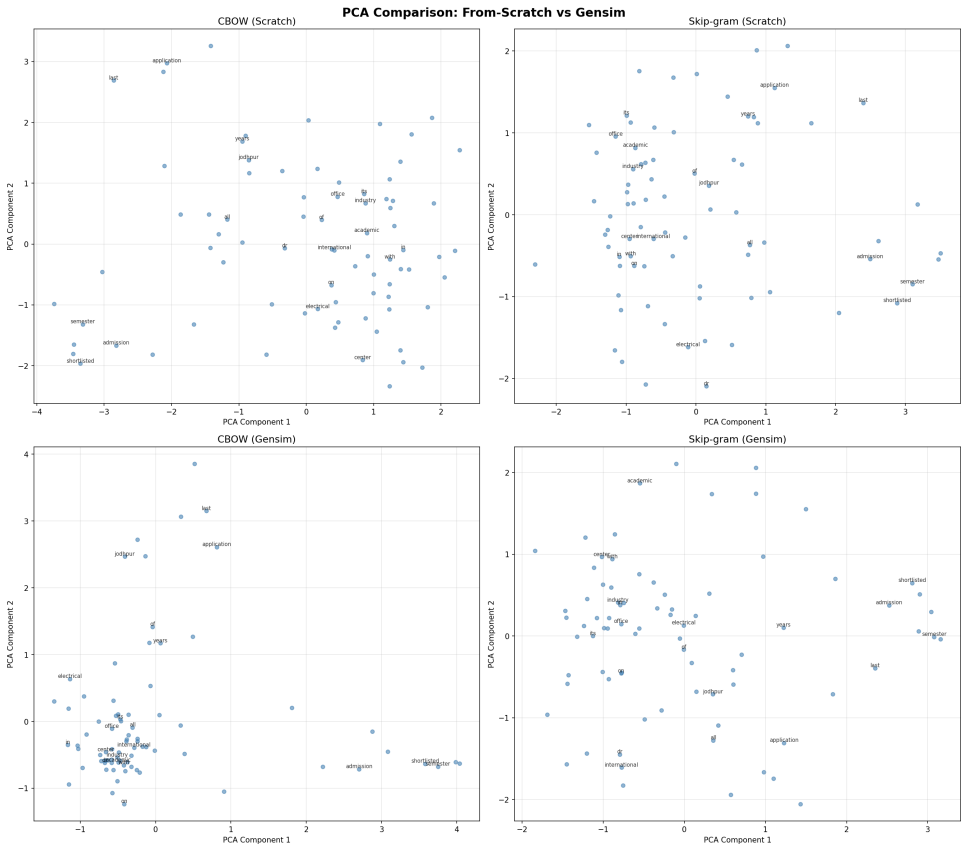


Figure 8: Comparison PCA

PROBLEM 2: CHARACTER-LEVEL NAME GENERATION USING RNN VARIANTS

TASK 0: DATASET

A dataset of 1,000 Indian names (First Name + Surname) was collected using an LLM to serve as the training source. The character-level vocabulary consists of 54 unique characters.

TASK 1: MODEL IMPLEMENTATION

I implemented and trained three recurrent neural network architectures. All models share a baseline configuration: Embedding dim = 32, Hidden size = 128, Learning rate = 0.003, and Batch size = 64.

1. **Vanilla Recurrent Neural Network (RNN)**
   - **Architecture:** An embedding layer followed by a standard RNN cell, outputting through a linear layer mapping hidden states to vocabulary logits.
   - **Trainable parameters:** 29,302
2. **Bidirectional Long Short-Term Memory (BLSTM)**
   - **Architecture:** Employs forward and backward LSTM cells. The outputs from both directions are concatenated before being passed to the final linear layer. Since it uses LSTMs, the gating mechanisms vastly increase parameter count.

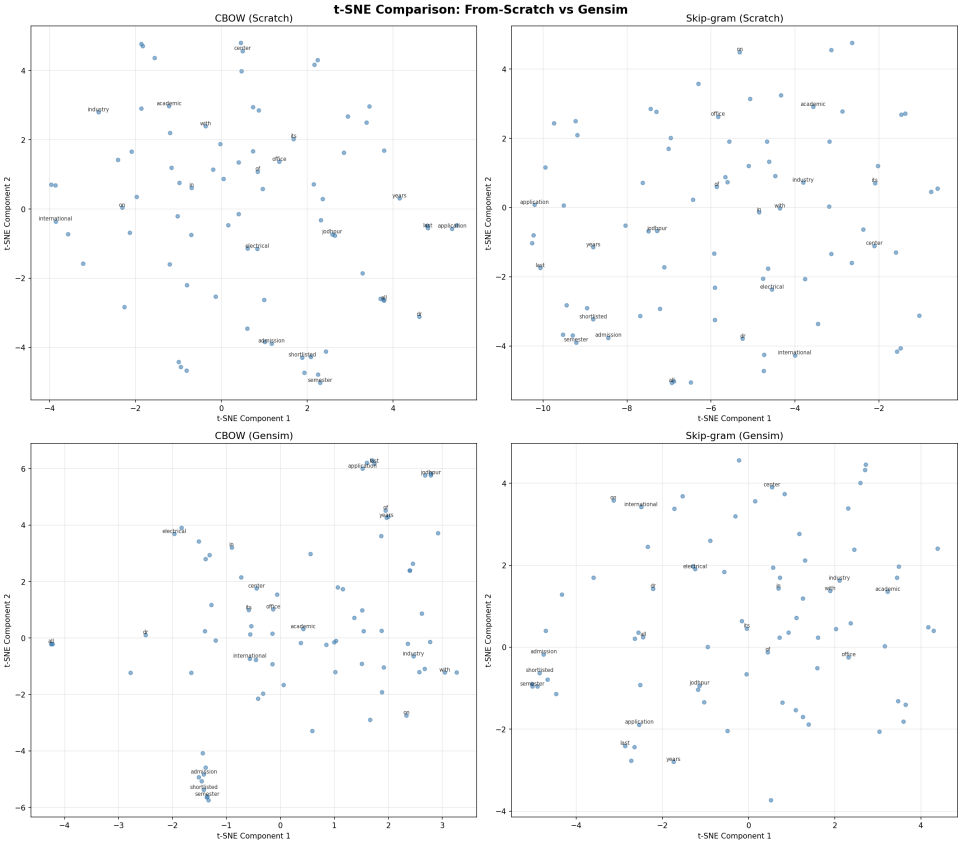


Figure 9: Comparison t-SNE

- **Trainable parameters:** 180,470

3. **RNN with Basic Attention Mechanism**

   - **Architecture:** Features a standard RNN cell combined with a learned attention mechanism. The attention weights help the model focus on the relevant past characters when deciding the next character.
   - **Trainable parameters:** 69,046

TASK 2: QUANTITATIVE EVALUATION

For each model, I generated 200 names and computed Novelty Rate (percentage of generated names not appearing in the training set) and Diversity (number of unique generated names divided by the total).

Model	Novelty %	Diversity
Vanilla RNN	77.5%	0.9950
BLSTM	100.0%	1.0000
RNN + Attention	100.0%	0.7400

Table 1: Quantitative Evaluation of RNN Models

Comparison: The Vanilla RNN exhibited high memorization with only a 77.5% novelty rate. The BLSTM and RNN+Attention models successfully learned the generative patterns without copying the dataset exactly (100% novelty). The BLSTM produced completely unique outputs (Diversity: 1.0), whereas the Attention model showed slight repetition (Diversity: 0.74).

TASK 3: QUALITATIVE ANALYSIS

Realism of generated names and failure modes:

- **Vanilla RNN:** Produced many highly realistic names like `Avantika Mishra`, `Kamran Patel`, and `Tagaan Shah`. These closely resembled actual human names, mostly because of its lower novelty rate (it memorized portions of the dataset). The main failure mode was occasionally producing long strings or merging names improperly.
- **BLSTM:** Showed significant failure in string coherency. Typical outputs look like `Uns Tey`, `Acyu Mel`. It struggled to produce standard recognizable phonemes correctly, likely because the text length is very short and bidirectional contexts are better suited for encoding full sentences rather than autoregressive sequential generation.
- **RNN + Attention:** Produced partially structured outputs with repeating consonants like `HomB Kut`, `Kogu Chel`, `KutBrt Gubhe`. It generated names with spaces and plausible short strings but failed to maintain the phonetic flow of an Indian name over multiple syllables.

Here is the terminal screenshot showing the entire training process and samples for Problem 2:

```
Loading training names...
  Loaded 1000 names

Building character vocabulary...
  Character vocabulary size: 54
  Characters: [' ', 'A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'R', 'S', 'T', 'U', 'V', 'W', 'Y', 'Z', 'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x'

==================================================
SHARED TRAINING HYPERPARAMETERS
==================================================
  EmbedDim dimension:  32
  Hidden size:         128
  Learning rate:       0.003
  Batch size:          64
  Epochs:              100
  Temperature (gen):   0.8
  Device:              cpu

==================================================
TRAINING: Vanilla RNN
==================================================

==================================================
MODEL ARCHITECTURE: Vanilla RNN
==================================================

  Layer Details:
  ----------------------------------------------
    embedding.weight                 → [54, 32]          [1,728 params]
    rnn_cell.0.W_ih.weight           → [128, 32]         [4,096 params]
    rnn_cell.0.W_ih.bias             → [128]             [128 params]
    rnn_cell.0.W_hh.weight           → [128, 128]        [16,384 params]
    fc_out.weight                    → [54, 128]         [6,912 params]
    fc_out.bias                      → [54]              [54 params]

  ==========================================
  Total Trainable Parameters: 29,302
  ==========================================

  Hyperparameters:
    Vocabulary size:   54
    Embedding dim:     32
    Hidden size:       128
    Number of layers:  1

  Epoch   1/100 — Loss: 3.1745
  Epoch  10/100 — Loss: 1.7696
  Epoch  20/100 — Loss: 1.5148
  Epoch  30/100 — Loss: 1.3038
  Epoch  40/100 — Loss: 1.1418
  Epoch  50/100 — Loss: 1.0302
  Epoch  60/100 — Loss: 0.9230
  Epoch  70/100 — Loss: 0.8481
  Epoch  80/100 — Loss: 0.8162
  Epoch  90/100 — Loss: 0.7725
  Epoch 100/100 — Loss: 0.7188
  Training complete!

Generating 200 names from Vanilla RNN...
  Successfully generated 200 names

==================================================
QUANTITATIVE EVALUATION: Vanilla RNN
==================================================
  Total names generated:    200
  Novel names (not in train): 147 / 200
  Novelty Rate:             73.5%
  Unique names:             192 / 200
  Diversity:                0.9600

==================================================
QUALITATIVE ANALYSIS: Vanilla RNN
==================================================

  Representative Generated Names (20 samples):
  ----------------------------------------------
    1. Padmaprisyattiranikar
    2. Aadhira Reddd
    3. Nikisha Tajaram
    4. Pankaj Kumar
    5. Tanaj Gambhra
    6. Nandana Dev
    7. Vaasha Jain
    8. Aapar Panda
    9. Mandakini Rajan
   10. Gorlit Sharma
   11. Kanchan Prakash
   12. Madhav Shetn
   13. Shekhar Kroogar
   14. Deepa Sinha
   15. Jasoda Patel
   16. Tumitt Sharma
   17. Avnish Saxena
   18. Ansh Dayanaayra
   19. Diksha Bashenshka Be
   20. Subham Mishra

  Length Statistics:
    Average length:  13.5 characters
    Min length:      5 characters
    Max length:      28 characters

  Common Failure Modes:
    Names too short (<3 chars): 0
    Names too long (>25 chars): 1
      Examples: ['SiddhartOS Mrilloni Mograjet']
    Names with space (full names): 198

  Top 10 Most Frequent Characters:
    'a': 365
    'h': 231
    ' ': 201
    'r': 192
    'i': 187
    'n': 177
    's': 161
    't': 108
    'e': 106
    'k': 103

  Generated names saved to: /home/jaishankar/Documents/NLU_Assign2/generated_vanilla_rnn.txt

==================================================
TRAINING: BLSTM
==================================================

==================================================
MODEL ARCHITECTURE: BLSTM
==================================================

  Layer Details:
  ----------------------------------------------
    embedding.weight                 → [54, 32]          [1,728 params]
    forward_cell.W_combined.weight   → [512, 160]        [81,920 params]
    forward_cell.W_combined.bias     → [512]             [512 params]
    backward_cell.W_combined.weight  → [512, 160]        [81,920 params]
    backward_cell.W_combined.bias    → [512]             [512 params]
    fc_out.weight                    → [54, 256]         [13,824 params]
    fc_out.bias                      → [54]              [54 params]

  ==========================================
  Total Trainable Parameters: 180,470
  ==========================================

  Hyperparameters:
    Vocabulary size:   54
    Embedding dim:     32
    Hidden size:       128

  Epoch   1/100 — Loss: 3.1160
  Epoch  10/100 — Loss: 0.0370
  Epoch  20/100 — Loss: 0.0076
  Epoch  30/100 — Loss: 0.0034
  Epoch  40/100 — Loss: 0.0021
  Epoch  50/100 — Loss: 0.0014
  Epoch  60/100 — Loss: 0.0010
  Epoch  70/100 — Loss: 0.0007
  Epoch  80/100 — Loss: 0.0006
  Epoch  90/100 — Loss: 0.0004
  Epoch 100/100 — Loss: 0.0004
  Training complete!

Generating 200 names from BLSTM...
  Successfully generated 200 names

==================================================
QUANTITATIVE EVALUATION: BLSTM
==================================================
  Total names generated:    200
  Novel names (not in train): 200 / 200
  Novelty Rate:             100.0%
  Unique names:             200 / 200
  Diversity:                1.0000

==================================================
QUALITATIVE ANALYSIS: BLSTM
==================================================

  Representative Generated Names (20 samples):
  ----------------------------------------------
    1. Vok Gzg
    2. Ava Twok
    3. Thbtrw Wip
    4. Qlrv Az
    5. Ir
    6. Arc Obtw
    7. Ftbl Bunn
    8. Exbm Cix
    9. Ocx Wwx
   10. W Ilprn
   11. Wdlab Rl
   12. Fexy Dtry
   13. F Dwsle
   14. Fqbr Sq
   15. Fnjut Awfl
   16. Zxwl Dqgdy
   17. Zdby Rg
   18. Sxyn Dxngp
   19. Zny Cnkus
   20. Pibq Nwg

  Length Statistics:
    Average length:  8.5 characters
    Min length:      4 characters
    Max length:      14 characters

  Common Failure Modes:
    Names too short (<3 chars): 4
      Examples: ['Ir', 'S', 'Ex', 'W']
    Names too long (>25 chars): 1
    Names with space (full names): 194

  Top 10 Most Frequent Characters:
    ' ': 215
    'x': 135
    'y': 111
    'w': 101
    'n': 94
    'b': 89
    'g': 86
    'c': 86
    'p': 83
    't': 72

  Generated names saved to: /home/jaishankar/Documents/NLU_Assign2/generated_blstm.txt

==================================================
TRAINING: RNN + Attention
==================================================

==================================================
MODEL ARCHITECTURE: RNN + Attention
==================================================

  Layer Details:
  ----------------------------------------------
    embedding.weight                 → [54, 32]          [1,728 params]
    rnn_cell.W_ih.weight             → [128, 160]        [20,480 params]
    rnn_cell.W_ih.bias               → [128]             [128 params]
    rnn_cell.W_hh.weight             → [128, 128]        [16,384 params]
    W_query.weight                   → [64, 128]         [8,192 params]
    W_key.weight                     → [64, 128]         [8,192 params]
    v_attn.weight                    → [1, 64]           [64 params]
    fc_out.weight                    → [54, 256]         [13,824 params]
    fc_out.bias                      → [54]              [54 params]

  ==========================================
  Total Trainable Parameters: 69,046
  ==========================================

  Hyperparameters:
    Vocabulary size:   54
    Embedding dim:     32
    Hidden size:       128
    Attention size:    64

  Epoch   1/100 — Loss: 3.0705
  Epoch  10/100 — Loss: 1.7999
  Epoch  20/100 — Loss: 1.4628
  Epoch  30/100 — Loss: 1.2429
  Epoch  40/100 — Loss: 1.1137
  Epoch  50/100 — Loss: 1.0102
  Epoch  60/100 — Loss: 0.9329
  Epoch  70/100 — Loss: 0.8746
  Epoch  80/100 — Loss: 0.8180
  Epoch  90/100 — Loss: 0.7761
  Epoch 100/100 — Loss: 0.7869
  Training complete!

Generating 200 names from RNN + Attention...
  Successfully generated 200 names

==================================================
QUANTITATIVE EVALUATION: RNN + Attention
==================================================
  Total names generated:    200
  Novel names (not in train): 200 / 200
  Novelty Rate:             100.0%
  Unique names:             166 / 200
  Diversity:                0.8300

==================================================
QUALITATIVE ANALYSIS: RNN + Attention
==================================================

  Representative Generated Names (20 samples):
  ----------------------------------------------
    1. Ann Rhh
```